The first law

antocciphysics | TEACHER KEY | CPA

Learning target: Use a consistent sign convention in an energy balance.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 4; 3 course days later.*
$Q=mL=1500$ J.

02. Explain a worked example (worked)
90 J in, 30 J out as work, 60 J increase; signs follow stated convention.

03. Look and explain (model)
$\Delta U=100-40=60$ J.

04. Build the solution (practice)
60 J increase.

05. Use the idea (reason)
Must know whether the system does work or has work done on it; the energy effect differs.

06. Choose a method (mixed) *Review: Unit 4, Day 17; 12 course days later.*
Inelastic; 4 J can become deformation, thermal energy or sound. Total energy is conserved.

07. Exit check (exit)
+20 J.

08. STRETCH: Apply it another way (stretch)
Examples 60 in and 20 out; 100 in and 60 out. Difference 40 J.

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01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 4; 3 course days later.*
$L=|Q|/m \approx 3.333\times10^{5}$ J/kg; sample Q=-100000 J.

02. Explain a worked example (worked)
90 J in, 30 J out as work, 60 J increase; signs follow stated convention.

03. Represent and interpret (model)
$\Delta U=100-40=60$ J.

04. Solve and justify (practice)
Q=-50 J; W_{by}=-60 J; $\Delta U=+10$ J.

05. Analyze the reasoning (reason)
Must know whether the system does work or has work done on it; the energy effect differs.

06. Choose a method (mixed) *Review: Unit 4, Day 17; 12 course days later.*
Momentum 48 kg·m/s before and after. K_i=288 J; K_f=148 J. Inelastic; 140 J leaves kinetic energy.

07. Exit check (exit)
Q=10 J; enters.

08. STRETCH: Apply it another way (stretch)
Q=-20, W=-50,$\Delta U=+30$ J. Reversal gives Q=+20, W=+50,$\Delta U=-30$ J.